GOVERNMENT POLYTECHNIC, VAISHALI

VAISHALI, BIHAR

A

Project Report On

"INSTITUTE MANAGEMENT SYSTEM APPLICATION"

Submitted in partial fulfillment of the requirement for the award of the Diploma in Computer Science Engineering

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CERTIFICATE

This is to certify that the minor project work entitled

"INSTITUTE MANAGEMENT SYSTEM APPLICATION"

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in partial fulfillment of the requirement for the award of Diploma in

Computer Science & Engineering

During the academic session **2023–2026**

Under the guidance of **Prof. Anmol Sir**, and it is submitted to **SBTE**.

Principal **Head of Department** **Guide**

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CERTIFICATE BY GUIDE

This is to certify that the project titled **"INSTITUTE MANAGEMENT SYSTEM APPLICATION"** is the bonafide work carried out by **ABHIRAJ KUMAR, ABHISHEK KUMAR, ATUL KUMAR, GAUTAM KUMAR, NIRAJ KUMAR,** students of Diploma in Computer Science & Engineering of Government Polytechnic Vaishali, affiliated to SBTE, during the academic session 2023-2026, in partial fulfillment of the requirement for the award of Diploma in Computer Science & Engineering.

I have verified the work for its documentation against problem statement and results presented in the report.

Signature of the Guide

Prof. Anmol Sir

Place: _____

Date: _____

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ABSTRACT

The **Institute Management System Application** is designed to manage and organize the various activities of an educational institute in an efficient and digital way. The main purpose of this system is to reduce manual work and improve the management of student and institute data.

This application helps in managing important information such as student details, course information, attendance records, notices and results. The system provides an easy interface for the administrator, teachers and students to access and manage the data.

By using this system, the institute can store and retrieve information quickly and securely. It helps in reducing paperwork and saves time for both staff and students. The application also improves communication between students and the institute administration.

The main goal of this project is to develop a simple and effective system that makes the management process easier, faster and more reliable for educational institutions.

1 About Project

The **Institute Management System Application** is developed to manage various activities of an educational institute in a digital way. In many institutes, student information, attendance, notices and results are handled manually which takes a lot of time and effort.

This project provides an online platform where the institute administration can manage student records, attendance and other important information easily. The system helps in reducing manual work and improves the efficiency of the management process.

The application is designed using modern technologies such as **XML for front-end design, Java for back-end processing and databases like Firebase, Supabase and SQLite for data storage**.

The main aim of this project is to create a simple and efficient system that can help the institute manage its activities in an organized and secure manner

1.1 Introduction

In the modern world, technology plays a very important role in managing information efficiently. Educational institutes also need digital systems to manage their daily activities such as student records, attendance, courses, and notices.

The **Institute Management System Application** is a software system designed to manage the activities of an educational institute in an organized and efficient manner. This system helps the institute administration to store, update and manage student and staff information easily.

Traditionally, many institutes maintain records manually using paper files, which is time-consuming and difficult to manage. The manual system also increases the chances of data loss and errors.

The Institute Management System provides a computerized solution to these problems. It helps in maintaining student details, course information, attendance records, results and other important data in a digital format.

This system improves the efficiency of the institute by reducing paperwork, saving time and providing quick access to information. It also helps administrators, teachers and students to access necessary information easily and securely.

The main aim of this project is to develop a simple and user-friendly application that helps in managing the overall activities of an institute in an effective way.

1.2 What is Institute Management System

An **Institute Management System** is a software application that helps educational institutions manage their daily activities and administrative tasks efficiently. It is designed to organize and maintain important information related to students, teachers, courses, attendance, results, and other academic records.

In traditional systems, most of the work in institutes is done manually using paper files and registers. This manual process is time-consuming, difficult to maintain and may lead to errors or loss of important data.

The Institute Management System provides a digital platform where all the information can be stored, updated, and accessed easily. It allows administrators and teachers to manage student records, attendance, and academic details in a systematic way.

This system also helps students to access important information such as notices, course details, schedules, and results. By using this application, the institute can improve efficiency, reduce paperwork, and save time.

The Institute Management System plays an important role in modern education management by providing a secure, organized and user-friendly way to manage institutional data. It ensures better communication between students, teachers and the institute administration.

Overall, the system helps in improving the performance and management of the institute by making all processes faster, easier and more reliable.

1.3 Problem Statement

In many educational institutes, most of the administrative work is still done manually using paper records and registers. Managing student details, attendance records, course information and results through manual methods is time-consuming and difficult to maintain.

The manual system also creates several problems such as data duplication, loss of records and difficulty in retrieving information quickly. When the number of students increases, it becomes more complicated to manage and organize all the data properly.

Another problem with the traditional system is that it requires a large amount of paperwork, which increases the chances of human errors and misplacement of important documents. It also takes more time to update and share information among students, teachers and the administration.

Therefore, there is a need for a computerized system that can manage institute activities efficiently. The **Institute Management System Application** is developed to solve these problems by providing a digital platform to store, manage and access information easily and securely.

This system helps in improving the efficiency, accuracy and speed of the institute management process.

1.4 Objective

The main objective of the Institute Management System Application is to develop a computerized system that helps in managing the various activities of an educational institute in an efficient and organized manner.

The system aims to reduce manual work and improve the management of student and institute data. It helps the institute administration to store, update and retrieve important information easily.

The objectives of this project are:

- To maintain student information in a digital format.
- To manage attendance records efficiently.
- To store and update course and subject details.
- To provide easy access to notices and announcements.
- To maintain student result and academic records.
- To reduce paperwork and manual effort.
- To improve communication between students, teachers and administration.

This system helps in making the institute management process faster, more accurate and reliable.

1.5 Proposed Solution

The proposed solution for the existing problems is to develop an **Institute Management System Application** that can manage the activities of an educational institute in a digital and efficient way.

This system will replace the traditional manual method of managing records with a computerized system. All important information such as student details, course information, attendance records, notices and results will be stored in a database.

The application will provide a user-friendly interface for administrators, teachers and students. The administrator will be able to add, update and delete records easily. Teachers will be able to manage attendance and academic information, while students will be able to view notices, course details and results.

The system will also provide fast data retrieval and secure data storage. By using this application, the institute can reduce paperwork, minimize errors and improve overall efficiency.

Thus, the Institute Management System Application will provide a reliable and effective solution for managing the various operations of an educational institute.

1.6Features

The **Institute Management System Application** provides several useful features that help in managing the activities of an educational institute in an efficient way. Some of the important features of the system are as follows:

1. Student Management
The system allows the administrator to add, update and manage student information such as name, course, contact details and academic records.

2. Course Management
The application helps in managing course and subject details of different departments in the institute.

3. Attendance Management
Teachers can easily record and maintain student attendance using the system.

4. Notice Management
The system allows the institute to publish important notices and announcements for students and staff.

5. Result Management
Student results and academic performance can be stored and managed in the system.

6. User-Friendly Interface
The application provides a simple and easy interface so that users can access and manage information without difficulty.

7. Data Security

The system stores information in a secure database, which helps in protecting important institute data.

8. Fast Data Access

Users can quickly search and retrieve information whenever required.

These features make the Institute Management System a useful and efficient tool for managing institute activities.

1.7Modules

The **Institute Management System Application** is divided into different modules to manage various activities of the institute efficiently. Each module performs a specific function in the system.

1. Admin Module

The Admin Module is responsible for managing the overall system. The administrator can add, update and delete student records, course details and other important information. The admin also controls user access and manages the entire system.

2. Student Module

The Student Module allows students to view their personal information, course details, attendance, notices and results. It helps students to access important academic information easily.

3. Teacher Module

The Teacher Module helps teachers to manage student attendance, update academic records and share important information related to courses and classes.

4. Attendance Module

This module is used to maintain and manage student attendance records. Teachers can mark attendance and the system stores it in the database for future reference.

5. Result Module

The Result Module helps in storing and displaying student results. Students can view their academic performance through the system.

6. Notice Module

The Notice Module allows the institute to publish important announcements, updates and notices for students and staff.

These modules work together to make the Institute Management System efficient, organized and easy to use.

2.1 Front End

Front End is the part of the application that interacts directly with the users. It is the visible part of the system where users can enter data, view information and perform different operations. In the Institute Management System Application, the front end provides a user-friendly interface for students, teachers and administrators to access the system.

The front end is responsible for displaying information and receiving input from users. It includes different forms, pages and menus that help users navigate through the system easily.

A good front-end design makes the application easy to use and improves the overall user experience. It allows users to interact with the system in a simple and efficient way.

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/gradiantmain"
    android:layout_marginTop="45dp"
    android:orientation="vertical">

    <!-- Include Toolbar -->
    <include
        android:id="@+id/toolbar"
        layout="@layout/drawerlayout" />

    <!-- Header Title -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="1st Year Engineering Branches"
        android:textColor="@color/white"
        android:textSize="20sp"
        android:textStyle="bold"
        android:padding="16dp"
        android:fontFamily="sans-serif-medium"
        android:gravity="center"
        android:background="#44000000" />

    <!-- Branch Selection Chips - Row 1 -->
    <HorizontalScrollView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```xml
        android:scrollbars="none"
        android:paddingVertical="8dp">

        <LinearLayout
            android:id="@+id/layoutSubjects1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:paddingHorizontal="12dp"
            android:paddingVertical="4dp">

            <androidx.cardview.widget.CardView
                android:layout_width="100dp"
                android:id="@+id/buttoncs1"
                android:layout_height="100dp"
                android:layout_margin="8dp"
                app:cardCornerRadius="16dp"
                app:cardElevation="6dp"
                app:cardBackgroundColor="@color/white">

                <LinearLayout
                    android:layout_width="match_parent"
                    android:layout_height="match_parent"
                    android:orientation="vertical"
                    android:gravity="center"
                    android:padding="8dp">

                    <ImageView
                        android:layout_width="48dp"
                        android:layout_height="48dp"
```

```
63                  android:src="@drawable/ic_cse"
64                  tools:ignore="ContentDescription" />
65
66              <TextView
67                  android:layout_width="wrap_content"
68                  android:layout_height="wrap_content"
69                  android:text="CS"
70                  android:textColor="@color/black"
71                  android:textSize="14sp"
72                  android:textStyle="bold"
73                  android:layout_marginTop="4dp" />
74
75              <TextView
76                  android:layout_width="wrap_content"
77                  android:layout_height="wrap_content"
78                  android:text="Computer Science"
79                  android:textColor="@color/gray"
80                  android:textSize="8sp"
81                  android:gravity="center"
82                  android:maxLines="1" />
83          </LinearLayout>
84      </androidx.cardview.widget.CardView>
85
86      <androidx.cardview.widget.CardView
87          android:layout_width="100dp"
88          android:layout_height="100dp"
89          android:id="@+id/buttonme1"
```

```
89          android:id="@+id/buttonme1"
90          android:layout_margin="8dp"
91          app:cardCornerRadius="16dp"
92          app:cardElevation="6dp"
93          app:cardBackgroundColor="@color/white">
94
95          <LinearLayout
96              android:layout_width="match_parent"
97              android:layout_height="match_parent"
98              android:orientation="vertical"
99              android:gravity="center"
100             android:padding="8dp">
101
102             <ImageView
103                 android:layout_width="48dp"
104                 android:layout_height="48dp"
105                 android:src="@drawable/ic_me"
106                 tools:ignore="ContentDescription" />
107
108             <TextView
109                 android:layout_width="wrap_content"
110                 android:layout_height="wrap_content"
111                 android:text="ME"
112                 android:textColor="@color/black"
113                 android:textSize="14sp"
114                 android:textStyle="bold"
115                 android:layout_marginTop="4dp" />
116
```

2.2 Architecture of Front-End User

The front-end architecture defines how the user interface of the system is designed and how users interact with the application.

In the Institute Management System Application, the front-end architecture includes different interfaces for users such as admin, teachers and students. Each user can access the system through a login interface and perform specific tasks based on their role.

The front-end consists of several components such as login pages, dashboard, data entry forms and information display pages. These components allow users to interact with the system and perform different operations.

For example, the admin can manage student records, teachers can update attendance and students can view their academic details. The front-end communicates with the back-end system to store and retrieve data from the database.
A well-designed front-end architecture helps in providing a smooth and efficient interaction between the users and the system.

2.3 Front-End Technology (XML)

XML (Extensible Markup Language) is used as the front-end technology to design the user interface of the application. It helps in creating the layout structure of the application and defines how different elements appear on the screen.

In Android applications, XML is commonly used to design layouts such as buttons, text views, edit texts, images and other interface components. It separates the design part from the program logic, which makes the application easier to manage and maintain.

Using XML, developers can organize the user interface in a structured way. It also helps in improving the readability of the code and allows easy modification of the layout without affecting the application logic.

In the **Institute Management System Application**, XML is used to design different user interface screens such as login page, registration page, dashboard and other forms used in the system.

Therefore, XML plays an important role in creating a simple, organized and user-friendly interface for the application.

3.1 Back End

Back End is the part of the application that works behind the user interface. It is responsible for processing data, storing information in the database and managing the overall functionality of the system.

In the **Institute Management System Application**, the back end handles all the important operations such as storing student details, managing course information, recording attendance and maintaining results.

The back-end system communicates with the front-end interface and processes the requests made by the users. When a user enters data through the front end, the back end stores that information in the database and retrieves it whenever required.

The back end ensures that all the data is stored securely and can be accessed efficiently. It plays a very important role in maintaining the reliability and performance of the application.

```java
package com.example.myapplication;

import ...

public class Semester1Activity extends AppCompatActivity {
    2 usages
    private CardView cardCs1, cardMe1, cardEc1, cardEe1, cardCv1, cardCtm1;
    3 usages
    private RecyclerView recyclerView;
    3 usages
    private PDFAdapter pdfAdapter;
    2 usages
    private Toolbar toolbar;
    4 usages
    private List<PDFModel> pdfList = new ArrayList<>();
    2 usages
    private SupabaseClient supabaseClient;
    6 usages
    private String currentSubject = "CS";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        EdgeToEdge.enable( $this$enableEdgeToEdge: this);
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_semester1);

        supabaseClient = SupabaseClient.getInstance( context: this);

        // Initialize views
        cardCs1 = findViewById(R.id.buttoncs1);
        cardMe1 = findViewById(R.id.buttonme1);
        cardEc1 = findViewById(R.id.buttonec1);
```

```java
        recyclerView = findViewById(R.id.recyclerView);
        recyclerView.setLayoutManager(new LinearLayoutManager( context: this));
        pdfAdapter = new PDFAdapter(pdfList,  isAdminMode: false);
        recyclerView.setAdapter(pdfAdapter);

        toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);

        if (getSupportActionBar() != null) {
            getSupportActionBar().setTitle("Contact Us");
            getSupportActionBar().setDisplayHomeAsUpEnabled(true);
            getSupportActionBar().setDisplayShowHomeEnabled(true);
        }

        // Set click listeners
        setClickListeners();

    }

    1 usage
    private void setClickListeners() {
        cardCs1.setOnClickListener( View v -> {
            currentSubject = "CS";
            loadPDFsBySubject("CS");
        });

        cardMe1.setOnClickListener( View v -> {
            currentSubject = "ME";
            loadPDFsBySubject("ME");
```

```
6 usages
93      private void loadPDFsBySubject(String subject) {
94          Toast.makeText( context: this,  text: "Loading " + subject + "...", Toast.LENGTH_SHORT).show();
95
96          supabaseClient.getPDFsBySemester( semester: "1", new SupabaseClient.PDFListCallback() {
                1 usage
97              @Override
98              public void onSuccess(List<PDFModel> pdfs) {
99                  runOnUiThread(() -> {
100                     pdfList.clear();
101                     for (PDFModel pdf : pdfs) {
102                         if (pdf.getSubject() != null && pdf.getSubject().equals(subject)) {
103                             pdfList.add(pdf);
104                         }
105                     }
106                     pdfAdapter.notifyDataSetChanged();
107
108                     if (pdfList.isEmpty()) {
109                         Toast.makeText( context: Semester1Activity.this,
110                                 text: "No content for " + subject, Toast.LENGTH_SHORT).show();
111                     }
112                 });
113             }
114
                no usages
115             @Override
116             public void onError(String error) {
117                 runOnUiThread(() ->
118                         Toast.makeText( context: Semester1Activity.this,
119                                 text: "Error: " + error, Toast.LENGTH_SHORT).show()
120                 );
```

```
115             @Override
116             public void onError(String error) {
117                 runOnUiThread(() ->
118                         Toast.makeText( context: Semester1Activity.this,
119                                 text: "Error: " + error, Toast.LENGTH_SHORT).show()
120                 );
121             }
122         });
123     }
124     @Override
125     public boolean onOptionsItemSelected(@NonNull MenuItem item) {
126         // Handle back button click
127         if (item.getItemId() == android.R.id.home) {
128             onBackPressed(); // This will call the default back button behavior
129             return true;
130         }
131         return super.onOptionsItemSelected(item);
132     }
133
134     @Override
135     public void onBackPressed() {
136         super.onBackPressed();
137         overridePendingTransition(android.R.anim.fade_in, android.R.anim.fade_out);
138     }
139 }
```

3.2 Architecture of Back-End User

The **Back-End Architecture** of the Institute Management System Application is responsible for handling the application logic, managing user requests and communicating with the database.

In this system, the back end is developed using **Java**, which processes the data received from the front-end interface. When a user performs an action such as login, registration, viewing data or updating information, the request is sent from the front end to the back-end system.

The back-end system then processes the request and communicates with the database to store or retrieve the required information. After processing the data, the back end sends the response back to the front end so that the user can view the result.

The back-end architecture also manages important operations such as user authentication, data processing and database communication.

In this project, different database technologies are used to manage data efficiently. **Firebase** is used for user authentication and password recovery, **Supabase** is used for content management and cloud data storage, and **SQLite** is used as a local database to store user details.

This architecture helps in ensuring secure data processing, fast data access and reliable communication between the user interface and the database.

3.3 Back-End Technology (Java)

Java is used as the back-end technology for developing the **Institute Management System Application**. It is a widely used programming language that helps in implementing the application logic and processing user requests.

In this project, Java is used to handle the functionality of the application such as user login, data processing and communication with the database. When a user

performs an action in the application, the request is processed by the Java code in the back end.

Java also helps in connecting the front-end interface with the database and ensures that the data is stored, updated and retrieved correctly. It allows the application to perform operations such as saving user details, retrieving information and managing different modules of the system.

Another advantage of using Java is that it provides strong security, reliability and platform independence. It helps developers create efficient and scalable applications.

Therefore, Java plays an important role in managing the internal working and overall functionality of the **Institute Management System Application**.

4 Database

A database is used to store and manage the data of the application. In the **Institute Management System Application**, different types of databases are used to manage user information and application data efficiently.

```java
package com.example.myapplication;

import ...

18 usages
public class SupabaseClient {
    6 usages
    private static final String SUPABASE_URL = "https://lqyktyczmrisiavnioqd.supabase.co";
    10 usages
    private static final String SUPABASE_ANON_KEY = "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOi
    3 usages
    private static final String BUCKET_NAME = "pdfs";
    3 usages
    private static final String TABLE_NAME = "pdfs";

    3 usages
    private static SupabaseClient instance;
    1 usage
    private Context context;

    1 usage
    private SupabaseClient(Context context) { this.context = context.getApplicationContext(); }

    public static synchronized SupabaseClient getInstance(Context context) {
        if (instance == null) {
            instance = new SupabaseClient(context);
        }
        return instance;
    }

    // Upload PDF using REST API
```

The databases help in storing, updating and retrieving data quickly and securely.

4.1 Firebase Database

Firebase is a cloud-based platform that provides various services for mobile and web applications. In this project, Firebase is used mainly for **user authentication and password management**.

Firebase allows users to create accounts, log in to the system and reset their passwords if they forget them. It provides a secure and reliable authentication system.

Another advantage of Firebase is that it works in real time and helps developers manage user accounts easily without creating complex server-side code.

Thus, Firebase helps in maintaining secure user authentication in the Institute Management System Application.

```java
public void uploadPDF(File pdfFile, String title, String subject,
                      String semester, UploadCallback callback) {
    new UploadPDFTask(callback).execute(pdfFile, title, subject, semester);
}

1 usage
private class UploadPDFTask extends AsyncTask<Object, Void, String> {
    5 usages
    private UploadCallback callback;
    3 usages
    private String errorMessage;

    1 usage
    UploadPDFTask(UploadCallback callback) { this.callback = callback; }

    @Override
    protected String doInBackground(Object... params) {
        HttpURLConnection connection = null;
        try {
            File pdfFile = (File) params[0];
            String title = (String) params[1];
            String subject = (String) params[2];
            String semester = (String) params[3];

            String fileName = UUID.randomUUID().toString() + "_" + pdfFile.getName();

            // Step 1: Upload file to storage
            String fileUrl = uploadFileToStorage(pdfFile, fileName);
            if (fileUrl == null) {
                throw new Exception("Failed to upload file to storage");
```

3.2 Supabase Database

Supabase is an open-source backend platform that provides database and storage services. In this project, Supabase is used for **content management and storing application data**.

It helps in managing information such as notices, course details and other important data required by the application. Supabase provides APIs that allow the application to access and manage data easily.

Supabase also ensures secure storage and efficient data management for the system.

```java
private String uploadFileToStorage(File file, String fileName) throws Exception {
    String boundary = "Boundary-" + UUID.randomUUID().toString();
    String lineEnd = "\r\n";
    String twoHyphens = "--";

    URL url = new URL( spec: SUPABASE_URL + "/storage/v1/object/" + BUCKET_NAME + "/" + fileName);
    HttpURLConnection connection = (HttpURLConnection) url.openConnection();
    connection.setRequestMethod("POST");
    connection.setDoOutput(true);
    connection.setDoInput(true);
    connection.setUseCaches(false);

    // Set headers
    connection.setRequestProperty("Authorization", "Bearer " + SUPABASE_ANON_KEY);
    connection.setRequestProperty("apikey", SUPABASE_ANON_KEY);
    connection.setRequestProperty("Content-Type", "multipart/form-data; boundary=" + boundary);

    // Write file data
    DataOutputStream outputStream = new DataOutputStream(connection.getOutputStream());

    // Start boundary
    outputStream.writeBytes( s: twoHyphens + boundary + lineEnd);
    outputStream.writeBytes( s: "Content-Disposition: form-data; name=\"file\"; filename=\"" + fileName + "\"" + lineEnd);
    outputStream.writeBytes( s: "Content-Type: application/pdf" + lineEnd);
    outputStream.writeBytes(lineEnd);

    // Write file content
    FileInputStream fileInputStream = new FileInputStream(file);
```

4.3 SQLite Database

SQLite is a lightweight and embedded database system that stores data locally within the application. It does not require a separate server and is easy to integrate with mobile applications.

In this project, SQLite is used to store **user details and other basic information locally** in the device. This helps in faster data access and allows the application to work even when internet connection is not available.

SQLite improves the performance of the application by providing quick and efficient data storage.

```java
public class DatabaseHelper extends SQLiteOpenHelper {

    1 usage
    private static final String DATABASE_NAME = "ProfileDB";
    1 usage
    private static final int DATABASE_VERSION = 1;
    8 usages
    private static final String TABLE_PROFILE = "profile";

    // Column names
    1 usage
    private static final String COLUMN_ID = "id";
    3 usages
    private static final String COLUMN_IMAGE = "image";
    3 usages
    private static final String COLUMN_NAME = "name";
    3 usages
    private static final String COLUMN_BRANCH = "branch";
    3 usages
    private static final String COLUMN_ROLL_NO = "roll_no";
    3 usages
    private static final String COLUMN_REG_NO = "reg_no";
    3 usages
    private static final String COLUMN_EMAIL = "email";

    // Create table query
    1 usage
    private static final String CREATE_TABLE = "CREATE TABLE " + TABLE_PROFILE + "("
            + COLUMN_ID + " INTEGER PRIMARY KEY AUTOINCREMENT,"
            + COLUMN_IMAGE + " BLOB,"
```

5 System Analysis

System Analysis is an important step in the development of a software system. It involves studying the existing system, identifying its problems and understanding the requirements for the new system.

In many educational institutes, most of the administrative work is done manually. Managing student records, attendance, course details and notices through paper-based methods takes a lot of time and effort.

System analysis helps in understanding how the current system works and what improvements are needed. Based on this analysis, a new system is designed that can perform the tasks more efficiently and accurately.

The **Institute Management System Application** is developed as a solution to manage the activities of the institute digitally and improve the overall efficiency of the management process.

5.1 Existing System

In the existing system, most of the institute activities are performed manually. Information related to students, courses, attendance and results is maintained using paper records and registers.

This method requires a lot of paperwork and makes it difficult to store and manage large amounts of data. When the number of students increases, it becomes more complicated to maintain records properly.

Searching and updating information in the manual system also takes more time and effort.

5.2 Limitations of Existing System

The existing manual system has several limitations:

- Large amount of paperwork
- Time-consuming process

- Difficulty in maintaining records
- High chances of human errors
- Risk of data loss or damage
- Difficulty in retrieving information quickly

These limitations make the manual system inefficient and less reliable.

5.3 Proposed System

The proposed system is the **Institute Management System Application**, which provides a computerized solution for managing institute activities.

In this system, all important information such as student details, attendance, notices and results are stored in a digital format using databases.

The system allows administrators, teachers and students to access the required information easily. It reduces manual work and improves the efficiency of the institute management process.

5.4 Advantages of Proposed System

The proposed system offers several advantages compared to the existing manual system:

- Reduces paperwork
- Saves time and effort
- Provides quick access to information
- Improves data accuracy
- Ensures secure data storage
- Makes the management process more efficient

Thus, the Institute Management System Application helps in improving the overall functioning of the institute.

6 Investigation Phase

The Investigation Phase is an important step in the system development process. In this phase, the existing system is studied and the requirements for the new system are identified.

During this phase, information about the current working system of the institute is collected and analyzed. The main objective of this phase is to understand the problems in the existing system and determine the needs of users.

In many institutes, most of the administrative tasks such as managing student records, attendance and notices are performed manually. This manual system requires a large amount of paperwork and takes more time to maintain records.

The investigation phase helps in identifying these problems and finding better solutions to improve the system. It involves studying the current processes, collecting user requirements and analyzing how the new system can solve the existing issues.

Based on the investigation, the Institute Management System Application is designed to provide a digital and efficient solution for managing institute activities. The new system helps in reducing manual work, improving accuracy and providing faster access to information.

Thus, the investigation phase plays an important role in understanding the system requirements and planning the development of a better and more efficient system.

7 System Design

System Design is the process of defining the structure, components and data flow of the system. It helps in understanding how the system will work and how different parts of the application interact with each other.

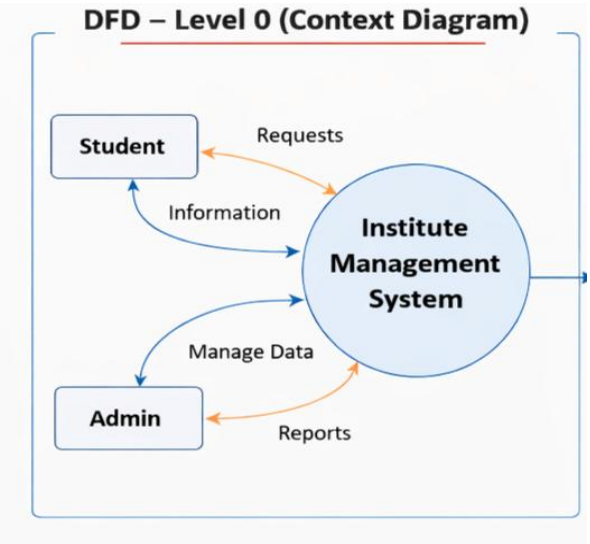
In the **Institute Management System Application**, the system design includes diagrams such as Data Flow Diagrams (DFD) and Entity Relationship Diagrams (ERD). These diagrams help in explaining the structure and working of the system in a clear and simple way.

7.1 Data Flow Diagram (Level 0)

DFD Level 0 is also called the **Context Diagram**. It shows the overall working of the system and the interaction between the users and the system.
Main entities involved in the system are:

- Student
- Admin
- Institute Management System

In this diagram, the student and admin interact with the Institute Management System to access or manage information such as student details, attendance and results.



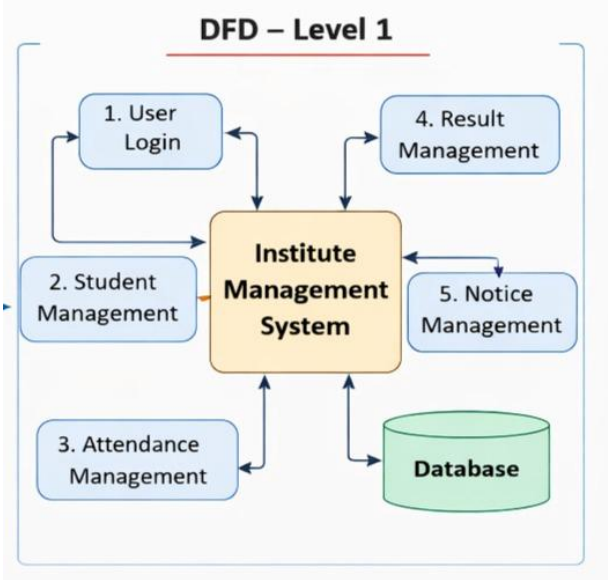
7.2 Data Flow Diagram (Level 1)

DFD Level 1 provides more details about the processes involved in the system. It divides the main system into smaller processes.

The main processes in the Institute Management System include:

- User Login
- Student Management
- Attendance Management
- Result Management
- Notice Management

Each process communicates with the database to store and retrieve information.



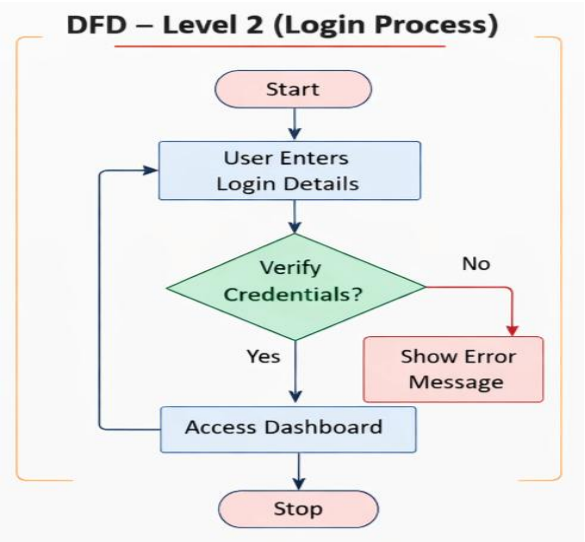
7.3 Data Flow Diagram (Level 2)

DFD Level 2 explains the processes of Level 1 in more detail. It shows how individual operations are performed within the system.

For example, in the login process:

- User enters login details
- System verifies the credentials
- If the details are correct, the user is allowed to access the dashboard

Similarly, other processes such as attendance management and result management are performed through detailed data flow operations.



7.4 ER Diagram (Entity Relationship Diagram)

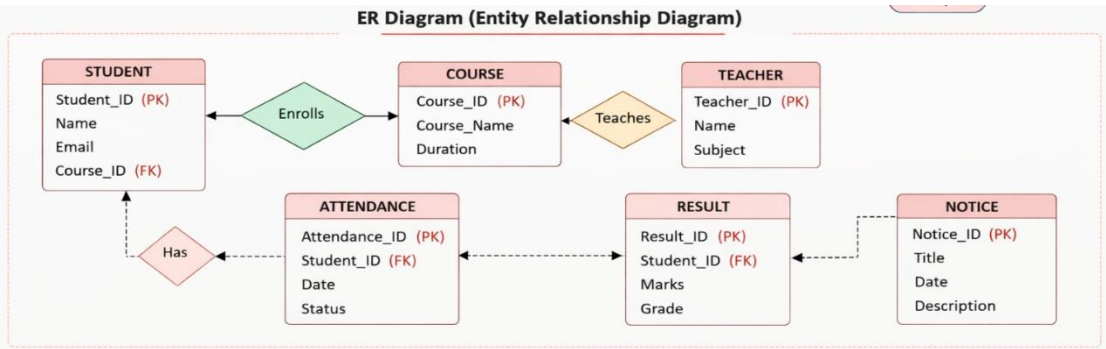
An **Entity Relationship Diagram (ER Diagram)** represents the relationship between different entities in the database.

In the Institute Management System, the main entities are:

- Student
- Teacher
- Course
- Attendance
- Result
- Notice

Each entity contains attributes such as student name, course details, attendance status and result information.

The ER diagram helps in designing the database structure and shows how different tables are related to each other.



8 Implementation

Implementation is the phase where the designed system is converted into a working application. In this stage, all modules of the **Institute Management System Application** are developed and integrated.

The application is implemented using **XML for the front-end interface and Java for the back-end processing**. The database technologies such as **Firebase, Supabase and SQLite** are used to store and manage application data.

Different modules such as login page, student dashboard, admin dashboard and attendance module are implemented to provide the required functionality of the system.

8.1 Login Page

The **Login Page** is the first screen of the application where users enter their login credentials to access the system.

In this module, users provide their email and password to log in to the application. The system verifies the user credentials using the authentication service provided by **Firebase**.

If the credentials are correct, the user is allowed to access the dashboard. If the credentials are incorrect, the system displays an error message.
This module ensures secure access to the application.

8.2 Student Dashboard

The **Student Dashboard** provides an interface where students can view important information related to their academic activities.
Through this dashboard, students can:

- View notices and announcements
- Check attendance details
- View result information
- Access other institute updates

The dashboard is designed to provide easy navigation and quick access to important information.

8.3 Admin Dashboard

The **Admin Dashboard** is used by the institute administrator to manage the system.

Through this dashboard, the administrator can:

- Add and manage student information
- Update attendance records
- Publish notices and announcements
- Manage academic data

The admin dashboard provides full control over the system and allows efficient management of institute activities.

8.4 Attendance Module

The **Attendance Module** is used to record and manage student attendance information.

In this module, the administrator or teacher can mark attendance for students. The attendance data is stored in the database and can be accessed later by students through the application.

This module helps the institute maintain accurate attendance records and makes the process faster and more organized.

9 Testing

Testing is an important phase in software development. It ensures that the application works correctly and all features perform as expected.

In the **Institute Management System Application**, different modules such as login, student dashboard, admin dashboard and attendance module were tested to check their functionality.

Testing helps in identifying errors and improving the performance of the system. After testing, necessary corrections are made to ensure that the application runs smoothly.

9.1 System Testing

System testing is performed to verify that the entire system works properly. All modules of the application are tested together to ensure that they interact correctly.

In this project, the system was tested to check:

- User login functionality
- Data storage and retrieval
- Dashboard operations
- Attendance management

The results showed that the system performs correctly and meets the requirements of the project.

9.2 Test Cases

Test cases are used to verify whether the application behaves as expected under different conditions.

Example test cases used in this project include:

Test Case	Input	Expected Result
Login Test	Correct email and password	User successfully logs in
Login Error	Incorrect password	Error message displayed
Attendance Entry	Admin marks attendance	Attendance stored in database
Notice View	Student opens notice	Notice displayed successfully

10 Advantages of the System

The Institute Management System Application provides several advantages:

- Reduces manual paperwork
- Saves time and effort
- Provides quick access to information
- Improves data accuracy
- Ensures secure data storage
- Makes institute management more efficient

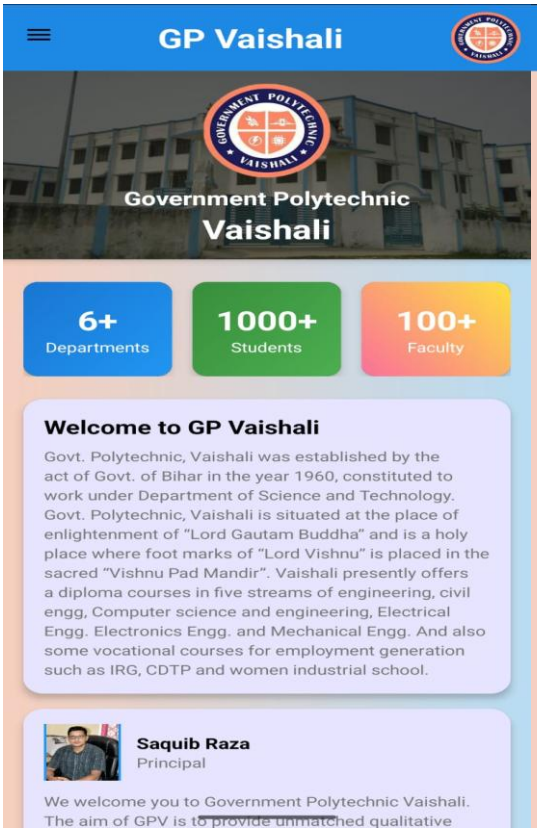
11 Limitations

Although the system provides many benefits, it also has some limitations:

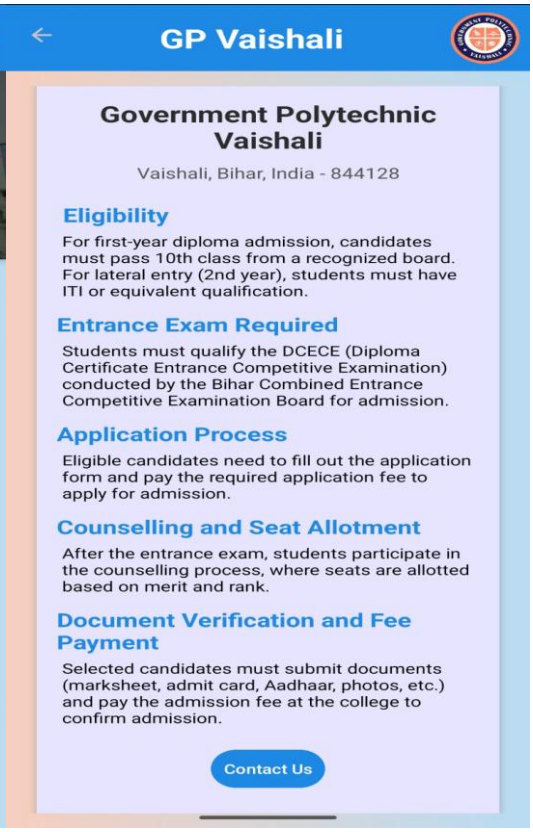
- Requires internet connection for some features
- Limited to basic institute management functions
- Needs regular updates and maintenance

These limitations can be improved in future versions of the application.

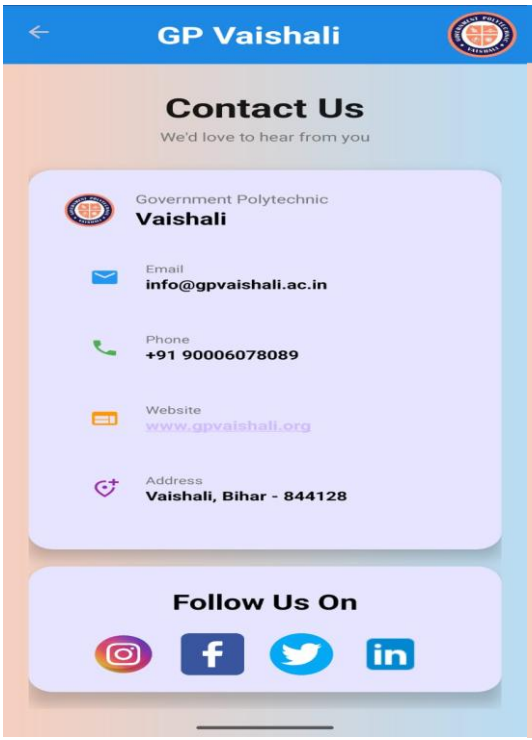
12 Output



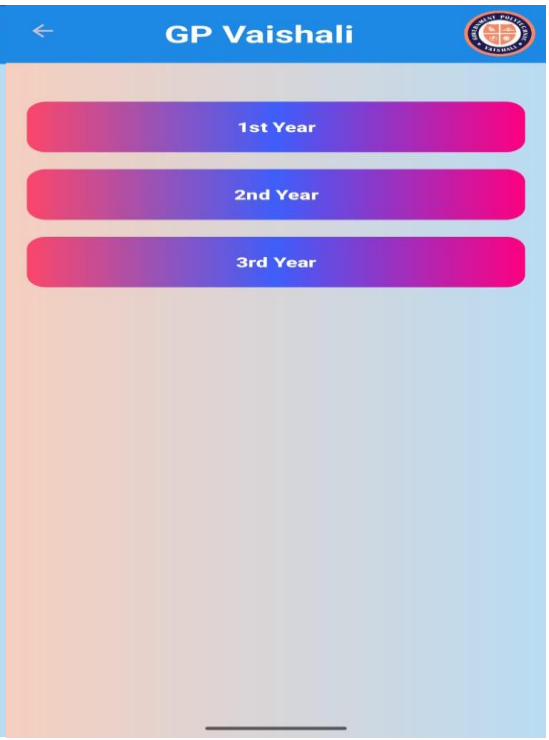
Home Page



Admission Process



Contact us



Academic Profile

13 Future Scope

The Institute Management System Application can be further improved in the future by adding more advanced features.

Possible future improvements include:

- Online fee payment system
- Online examination module
- Notification system for students
- Integration with other educational platforms

These improvements will make the system more useful and efficient for institutes.

14 Conclusion

The **Institute Management System Application** is designed to simplify and improve the management of institute activities.

The system provides a digital platform to manage student information, attendance and notices effectively. It reduces manual work and helps administrators manage data more efficiently.

The application demonstrates how modern technologies such as XML, Java and databases can be used to develop a useful and efficient management system.

Overall, the project successfully achieves its objective of providing a simple and effective institute management solution.